The University of Texas at Austin
Optimization

Homework 9 Solutions

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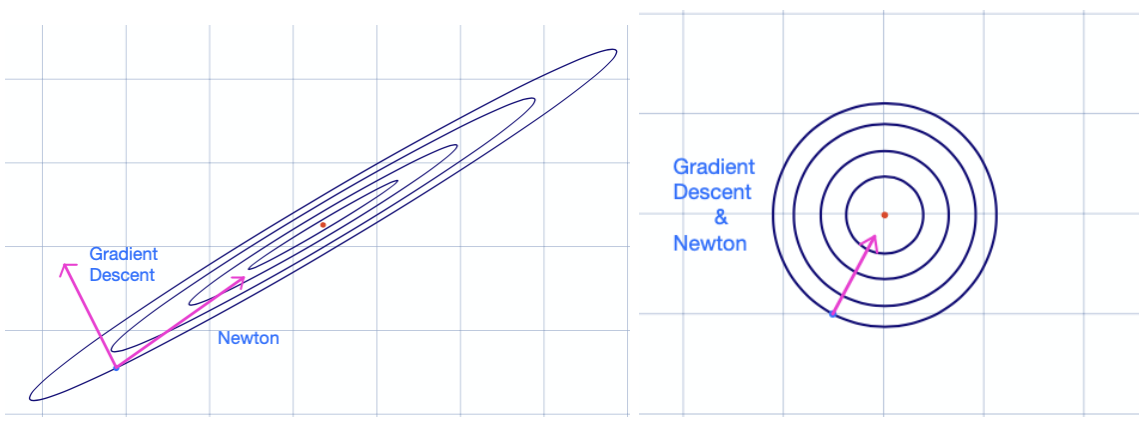
1. Consider the convex unconstrained problem:

$$\min: \ f(\boldsymbol{x}),$$

where $f(\boldsymbol{x})$ is a strongly convex function. One of our motivation for Newton's method was based on the idea of finding the "best" affine transformation possible at each step of a gradient-descent algorithm. We illustrate the key idea in Figure 1. Gradient descent, by its definition, moves along the negative gradient at the current point. The gradient (and hence negative gradient) always points in the direction (locally) perpendicular to the level set. In the first example, due to the elongation of the level sets, the gradient is pointing in a direction almost perpendicular to the direction of the optimal solution. In the second example, there is no elongation, and the level sets are spherical, centered on the optimal solution. Here, gradient descent points directly towards the optimal solution. We see that in both examples, Newton's method is not affected by the local elongation of the level sets.



(a) Gradient Descent and Newton updates can be very different – nearly orthogonal – in the event that the level sets are (locally) very elongated.

(b) Gradient Descent and Newton updates coincide in the setting where the level sets are (locally) spherical.

Figure 1: Newton vs Gradient Descent

You will explore this explicitly in this exercise and the next.

(a) Consider the ellipse given by:

$$\mathcal{E} = \left\{ \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : \boldsymbol{x}^\top Q \boldsymbol{x} \leq 1 \right\},$$

where

$$Q = \begin{bmatrix} 1/9 & 0 \\ 0 & 1 \end{bmatrix}.$$

Plot the ellipse $\mathcal{E}$.

Solution: *As shown in Figure 2, this is the ellipse with axes parallel to the coordinate axes, going through the points* $\{(3,0),(-3,0),(0,1),(0,-1)\}$.

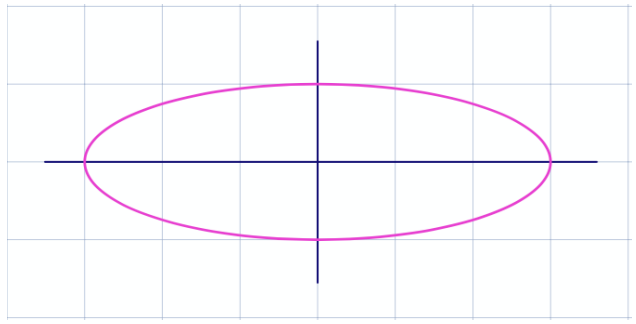


Figure 2: Elongated Axis-Parallel Ellipse

(b) Now consider the affine transformation (change of coordinates) given by

$$A\boldsymbol{y} = \boldsymbol{x},$$

where

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

Plot the ellipse in the new $\boldsymbol{y}$-coordinates.

Solution: *We have rescaled, so that now the ellipse is the unit circle, as shown in Figure 3.*

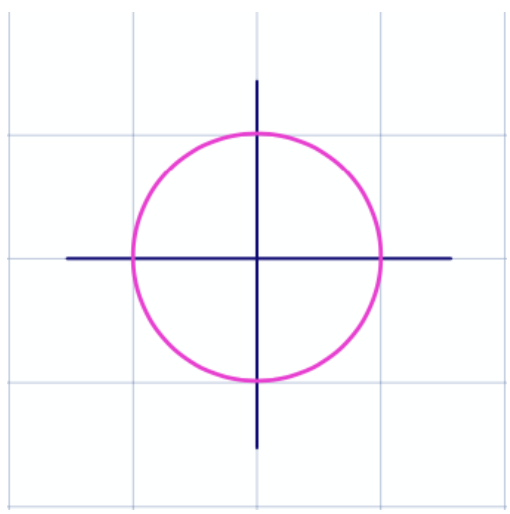


Figure 3: Rescsaled Ellipse

(c) Now consider the ellipse given by

$$\mathcal{E} = \left\{ \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : \boldsymbol{x}^\top Q\boldsymbol{x} \leq 1 \right\},$$

where

$$Q = \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}.$$

Plot the ellipse $\mathcal{E}$.

Solution: *As shown in Figure 4, this is the elongated and rotated ellipse, going through the points*

$$\{(1,0),(-1,0),(0,1),(0,-1),(\sqrt{5},-\sqrt{5}),(-\sqrt{5},\sqrt{5})\}.$$

We can obtain this directly by calculation. We know that the major and minor axes of the ellipse correspond to the eigenvectors of the matrix Q. These are $\boldsymbol{v}_1^\top = (1,1)$ and $\boldsymbol{v}_1^\top = (1,-1)$. Immediately, we see that the points $\{(1,0),(-1,0),(0,1),(0,-1)\}$ satisfy the equation $\boldsymbol{x}^\top Q\boldsymbol{x} = 1$. Since we know the elongation is in the $(1,-1)$ direction, we set $x_1 = -x_2 = \lambda$ and plug in, finding:

$$\lambda^2 - 1.8\lambda^2 + \lambda^2 = 1,$$

from which we find $\lambda = \sqrt{5}$.

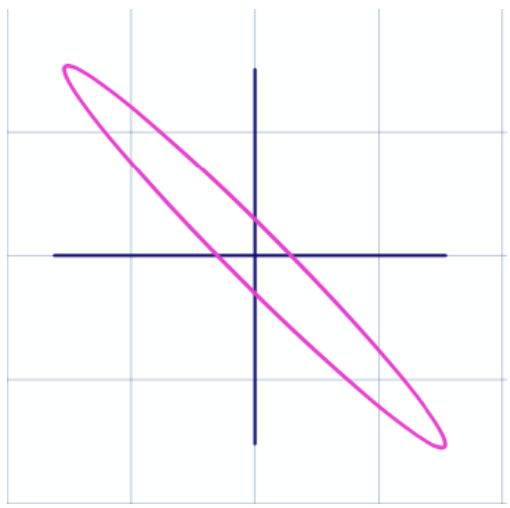


Figure 4: Elongated Rotated Ellipse

(d) Find a change of coordinates so that the ellipse in the new coordinates is a sphere. That is, find a matrix A so that the new ellipse in the $\boldsymbol{y}$-coordinates is spherical.

Solution: *The previous parts of this exercise suggest the solution. If we have $A\boldsymbol{y} = \boldsymbol{x}$, then in the $\boldsymbol{y}$ coordinates, the equation of the ellipse reads*

$$\boldsymbol{y}^\top A^\top Q A\boldsymbol{y} \leq 1.$$

Since Q is positive definite, it has a square root, which has an inverse. If we take $A = Q^{-1/2}$, then the ellipse indeed becomes simply the sphere:

$$\boldsymbol{y}^\top \boldsymbol{y} \leq 1.$$

Thus, specifically for our example here: letting λ_1 and λ_2 denote the eigenvalues of Q and $\boldsymbol{v}_1, \boldsymbol{v}_2$ its eigenvectors,

$$Q = \lambda_1 \boldsymbol{v}_1\boldsymbol{v}_1^\top + \lambda_2 \boldsymbol{v}_2\boldsymbol{v}_2^\top,$$

and

$$Q^{-\frac{1}{2}} = \frac{1}{\sqrt{\lambda_1}}\boldsymbol{v}_1\boldsymbol{v}_1^\top + \frac{1}{\sqrt{\lambda_2}}\boldsymbol{v}_2\boldsymbol{v}_2^\top.$$

For us, we have

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

and $\lambda_1 = 1.8$ *and* $\lambda_2 = 0.1$.

2. We will apply the same ideas to Gradient descent and Newton's method. Consider again a convex function $f(\cdot)$, and our standard unconstrained convex optimization problem.

   For A an invertible matrix, consider the change of coordinates $A\boldsymbol{y} = \boldsymbol{x}$, and accordingly, the function

   $$g(\boldsymbol{y}) = f(A\boldsymbol{y}).$$

   - Consider some starting point $\boldsymbol{x}_0$, and the sequence produced by gradient descent on $f(x)$ starting from $\boldsymbol{x}_0$, and using step-size η_k at iteration k. Define also $\boldsymbol{y}_0$ given by $A\boldsymbol{y}_0 = \boldsymbol{x}_0$, and consider the sequence produced by performing gradient descent on $g(\boldsymbol{y})$ starting from $\boldsymbol{y}_0$, and with the same step size. Show that in general these will not be the same by providing a specific example of a function $f(\cdot)$ and a matrix A, and demonstrating that the trajectories are not the same.
   - Now repeat this for Newton's method, where the updates $\{\boldsymbol{x}_k\}$ and $\{\boldsymbol{y}_k\}$ are generated by using *undamped* Newton's method on $f(\boldsymbol{x})$ and $g(\boldsymbol{y})$, respectively. Show that $A\boldsymbol{y}_k = \boldsymbol{x}_k$ for all k. Show this *in general*, not through example.

   Solution: $\nabla^2 g(y) = A^T\nabla^2 f(Ay)A$. *We only need to check if* $\Delta x = A\Delta y$. *That is,*

   $$\begin{aligned}\Delta y &= \nabla^2 g(y)^{-1}\nabla g(y) \\ &= A^{-1}\nabla^2 f(Ay)^{-1}A^{-T}A^T\nabla f(Ay) \\ &= A^{-1}\Delta x,\end{aligned}$$

   which proves the claim.

3. Consider the ℓ_1-regularized regression problem

   $$\min_{\boldsymbol{x}} \quad \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{x}\|_1 \tag{1}$$

   Where $\boldsymbol{x} \in \mathbb{R}^d$, $\boldsymbol{y} \in \mathbb{R}^n$. Show that a point $\bar{x}$ is an optimum of this problem if and only if there exists a $\boldsymbol{z} \in \mathbb{R}^d$ such that both the following hold:
   (a) $-A'(\boldsymbol{y} - A\bar{\boldsymbol{x}}) + \lambda\boldsymbol{z} = 0$
   (b) For every $i \in [d]$, $z_i = sign(\bar{x}_i)$ if $\bar{x}_i \neq 0$, and $|z_i| \leq 1$ if $\bar{x}_i = 0$.

   Solution: *The sub-gradient of the object function at* x *is*

   $$A^\top(A\bar{\boldsymbol{x}} - \boldsymbol{y}) + \lambda\boldsymbol{z},$$

   where $\boldsymbol{z} \in \partial(\|\boldsymbol{x}\|_1)$. *For sub-gradient of* l_1 *norm, it is easy to verify that if* $x_i \neq 0$, *then* $z_i = sign(x_i)$, *and otherwise* $|z_i| \leq 1$. *The optimality condition is* $A'(A\bar{\boldsymbol{x}} - \boldsymbol{y}) + \lambda\boldsymbol{z} = 0$ *for some* $\boldsymbol{z} \in \partial(||\bar{x}||_1)$, *as claimed.*

4. Consider the Lasso problem from the previous exercise: given a $n \times d$ matrix A, and a vector $\boldsymbol{y} \in \mathbb{R}^n$, and a positive scalar λ:

$$\min_{\boldsymbol{x}} : \ \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{x}\|_1.$$

As we have discussed extensively, this problem is convex. But as it is the sum of a smooth term ($\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2$) and a non-smooth term ($\|\boldsymbol{x}\|_1$) the overall objective is not smooth. We have spent some time developing a composite-optimization approach to solve this, namely, the proximal gradient method. The goal of the proximal gradient approach is to exploit the fact that the first term is smooth, and the second term has an "easy" to evaluate prox function. On the other hand, since it is convex, we could just use the sub-gradient method to solve directly, without treating the two terms separately.

(a) Write down the subgradient method for this problem. Be as explicit as possible when you write out the subgradient.

Solution: *We more or less have the solution from the previous exercise. The subgradient, like the gradient, is the sum of the subgradients of each component. Thus, as explained in the solution above, at a point $\boldsymbol{x}_t$, the subgradient is given by:*

$$\boldsymbol{g}_t = A^\top(A\boldsymbol{x}_t - \boldsymbol{y}) + \lambda\boldsymbol{z},$$

where $\boldsymbol{z} \in \partial(\|\boldsymbol{x}_t\|_1)$. For sub-gradient of l_1 norm, it is easy to verify that if $x_i \neq 0$, then $z_i = sign(x_i)$, and otherwise $|z_i| \leq 1$.

Given $\boldsymbol{g}_t$, we can now write the subgradient iteration:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \eta\boldsymbol{g}_t.$$

(b) Now write the proximal gradient update. Use a fixed step size, $\eta_t = \eta$. What is a good choice of η?

Solution: *Recall that for a composite optimization problem of the form:*

$$\min : \ g(\boldsymbol{x}) + h(\boldsymbol{x}),$$

the proximal gradient update is given by

$$\boldsymbol{x}_{t+1} = \mathrm{prox}_{\eta h}(\boldsymbol{x}_t - \eta\nabla g(\boldsymbol{x}_t)),$$

where a good choice of the step size η is the same fixed step size we would have chosen if we were running gradient descent on the smooth (by assumption) function $g(\cdot)$, namely, $\eta = 1/\beta$, for β the smoothness parameter of $g(\cdot)$.

In this case, writing out $g(\boldsymbol{x})$ we see that the leading term is quadratic of the form $\frac{1}{2}\boldsymbol{x}^\top A^\top A\boldsymbol{x}$, and hence the paramer β is given by $\lambda_{\max}(A^\top A)$, the largest eigenvalue of $A^\top A$.

The function h is $\lambda\|\cdot\|_1$, and hence its prox operator is the soft thresholding function we discussed in the lectures. Specifically, the soft thresholding operator at level α is given by:

$$\tau_\alpha(\boldsymbol{u})_i = (|u_i| - \alpha)_+\mathrm{sign}(u_i).$$

Thus, for $h(\boldsymbol{x}) = \lambda\|\boldsymbol{x}\|_1$,

$$\text{prox}_{\eta h}(\boldsymbol{u}) = \tau_{\lambda\eta}(\boldsymbol{u}),$$

and hence we finally have:

$$\boldsymbol{x}_{t+1} = \tau_{\lambda\eta}(\boldsymbol{x}_t - \eta\nabla g(\boldsymbol{x}_t)),$$

where $\eta = 1/\beta$*, as given above.*

5. For the same setting as above, generate random data, and solve the problem numerically, plotting the rate of convergence $\|\boldsymbol{x}_t - \boldsymbol{x}^*\|_2$. I recommend: choose $n = 50$ and $d = 200$, and let $\boldsymbol{x}^*$ a matrix with only 5 non-zero values (say, choose them to be -10 or 10). Choose the matrix A by selecting each of its $n \times d$ entries uniformly at random from a standard Gaussian distribution ($A_{ij} \sim N(0, 1)$). Then, let $\boldsymbol{e}$ be a random n-dimensional vector, where each entry is generated according to $N(0, 0.1)$. And finally, set

$$\boldsymbol{y} = A\boldsymbol{x}^* + \boldsymbol{e}.$$

The game now is to pretend we do not know $\boldsymbol{x}^*$, and to try to estimate it by solving the Lasso problem. That is,

$$\hat{\boldsymbol{x}} = \arg\min : \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2 + \lambda\|\boldsymbol{x}\|_1.$$

Use sub-gradient and prox grad method to solve this, and compare the rate of convergence for each method by plotting $\|\boldsymbol{x}_t - \boldsymbol{x}^*\|_2$ vs iteration.

Solution: *You should find here that the proximal gradient algorithm converges much faster than the subgradient method, as predicted by the theoretical results given in the lectures.*